Moderated and Nonlinear Models: Part 2.

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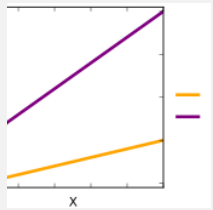
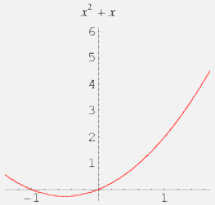
Definitions



Moderated: The effect of one variable changes as a function of another variable.



Nonlinear: The effect of one variable changes as a function of itself.

Nonlinear Models

- Nonlinear models comprise a wide category of models. This includes what are called curvilinear models and truly non-linear models.
  - Curvilinear model: $y_i = \beta_0 + \beta_1(x_1) + \beta_2(x_1^2) + \beta_3(x_1^3) + \varepsilon_i$
  - Truly non-linear model: $y_i = \alpha_1 - \gamma_1 e^{(-\delta_1/x_i)}$

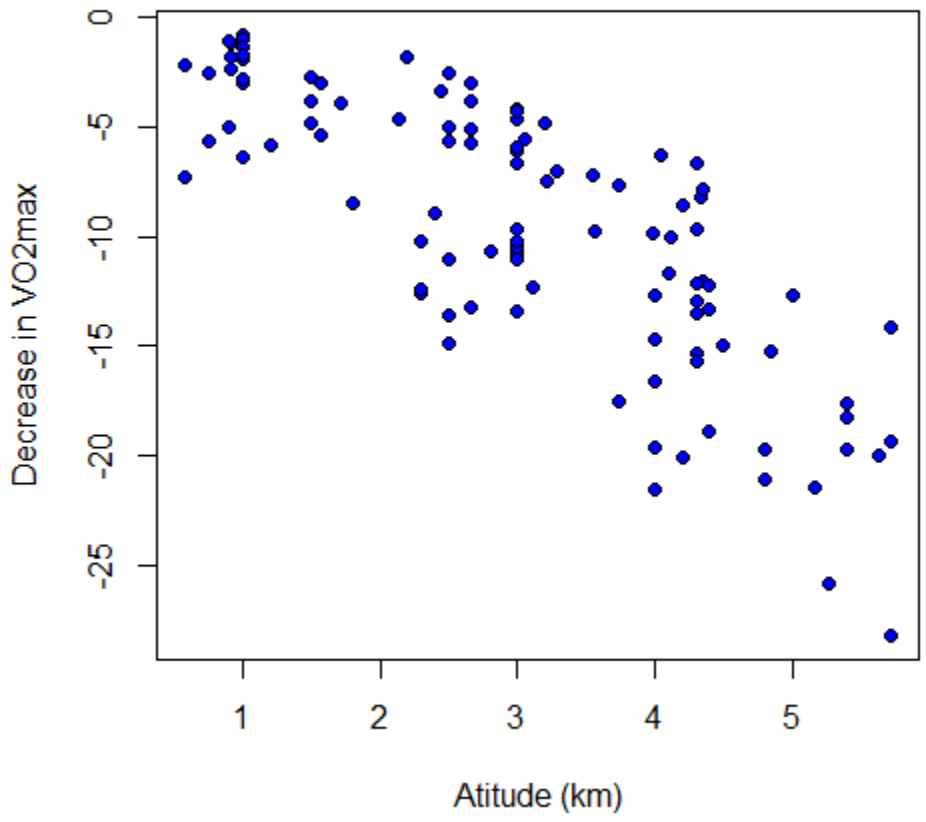
VO_{2max} changes as a function of altitude.

Using the ALTITUDE data, we can see how the VO_{2max} of 105 athletes changed as a function of the altitude at which their VO_{2max} was assessed.

If we wanted to see if the decrease changes as a linear function of altitude, that is something we are equipped to do:

MODEL A: $y_i = \beta_0 + \beta_1(ALT_i) + \epsilon_i$

MODEL C: $y_i = \beta_0 + \epsilon_i$



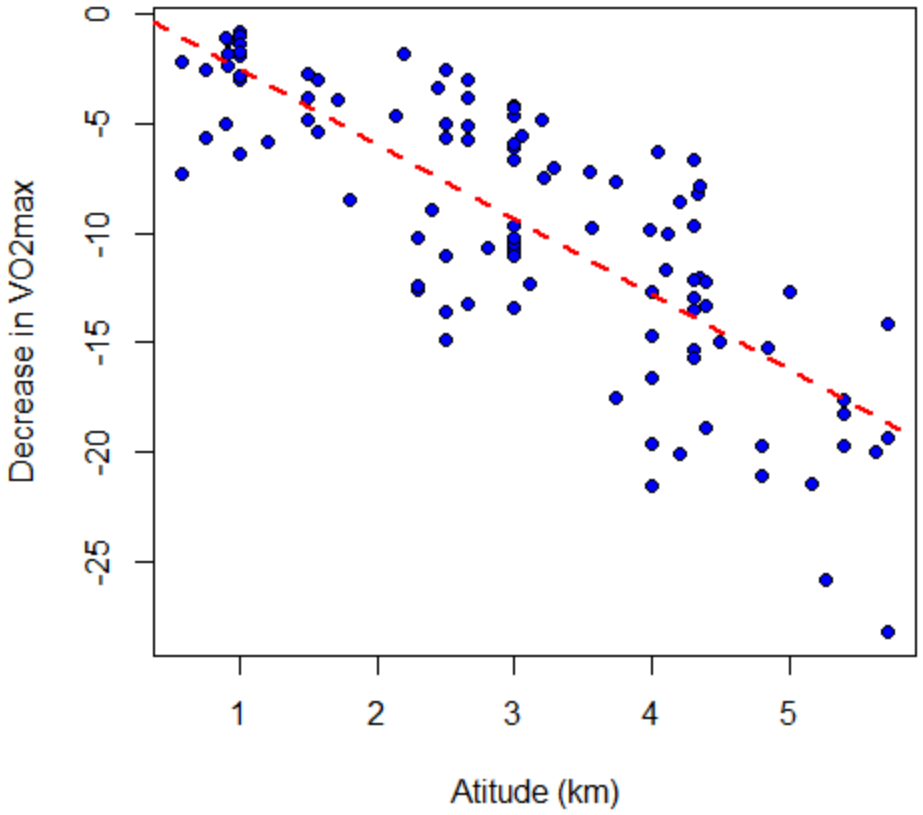
VO_{2max} changes as a function of altitude.

```
lm(formula = decrease ~ alt_km, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-9.5185 -2.1186  0.3748  2.9344  7.2001

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.9817     0.8855   1.109     0.27
alt_km       -3.4376     0.2616 -13.143   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.786 on 103 degrees of freedom
Multiple R-squared:  0.6264,    Adjusted R-squared:  0.6228
F-statistic: 172.7 on 1 and 103 DF,  p-value: < 2.2e-16
```



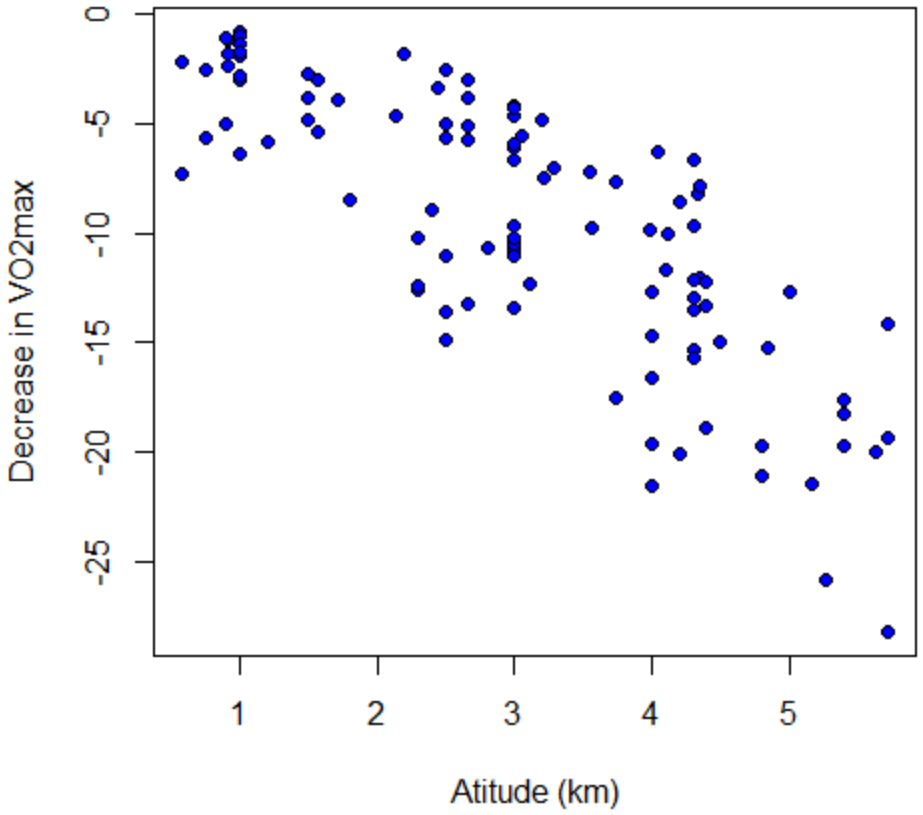
Is the VO_{2max} change curvilinear?

That is, is the change in VO_{2max} max constant?

Or is the difference between 1 and 2km different than the different between 4 and 5km?

MODEL A: $y_i = \beta_0 + \beta_1(ALT_i) + \beta_2(ALT_i^2) + \epsilon_i$

MODEL C: $y_i = \beta_0 + \epsilon_i$



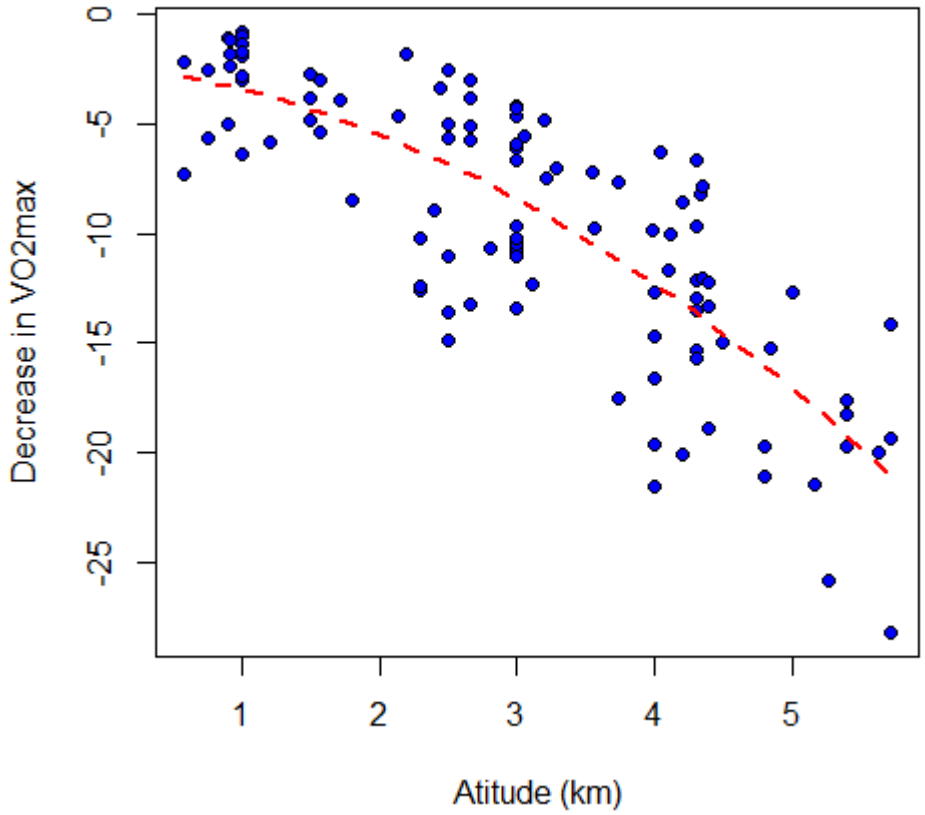
Is the VO_{2max} change curvilinear?

```
lm(formula = decrease ~ alt_km + alt_km_sq, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.3091 -2.4191  0.9296  2.4161  7.0271 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)  -2.2708     1.5327  -1.482   0.1415  
alt_km       -0.6921     1.0994  -0.630   0.5304  
alt_km_sq    -0.4532     0.1766  -2.567   0.0117 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.687 on 102 degrees of freedom
Multiple R-squared:  0.6491,	Adjusted R-squared:  0.6422 
F-statistic: 94.35 on 2 and 102 DF,  p-value: < 2.2e-16
```



Which models are being compared?

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How do we interpret the parameters?

$$y_i = -2.2708 - 0.6921(ALT_i) - 0.4532\left(ALT_i^2\right) + e_i$$

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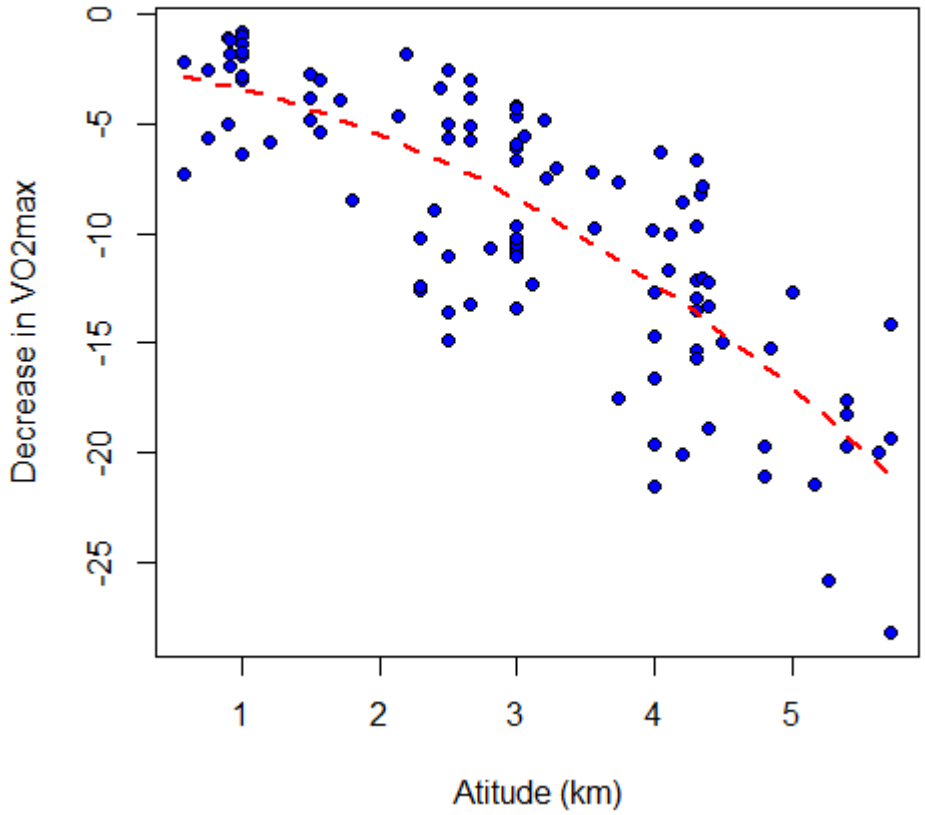

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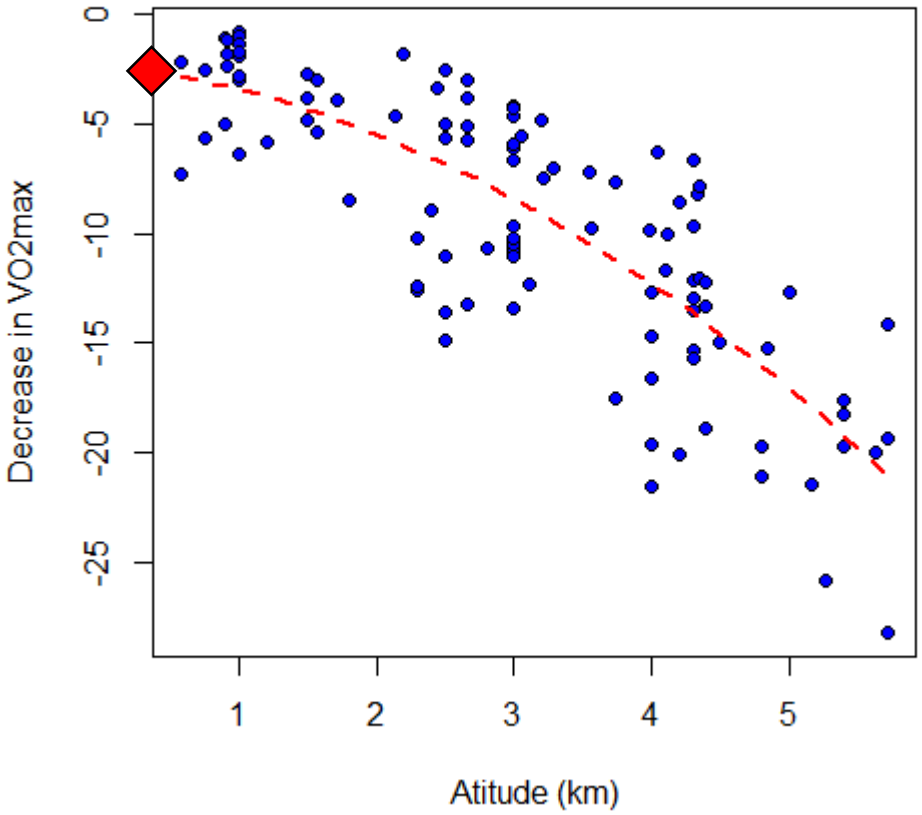
$$y_i = -2.2708 - 0.6921(ALT_i) - 0.4532\left(ALT_i^2\right) + e_i$$

$$y_i = -2.2708 - 0.6921(\mathbf{0}) - 0.4532(\mathbf{0}) + e_i$$

$$y_i = -2.2708 + e_i$$



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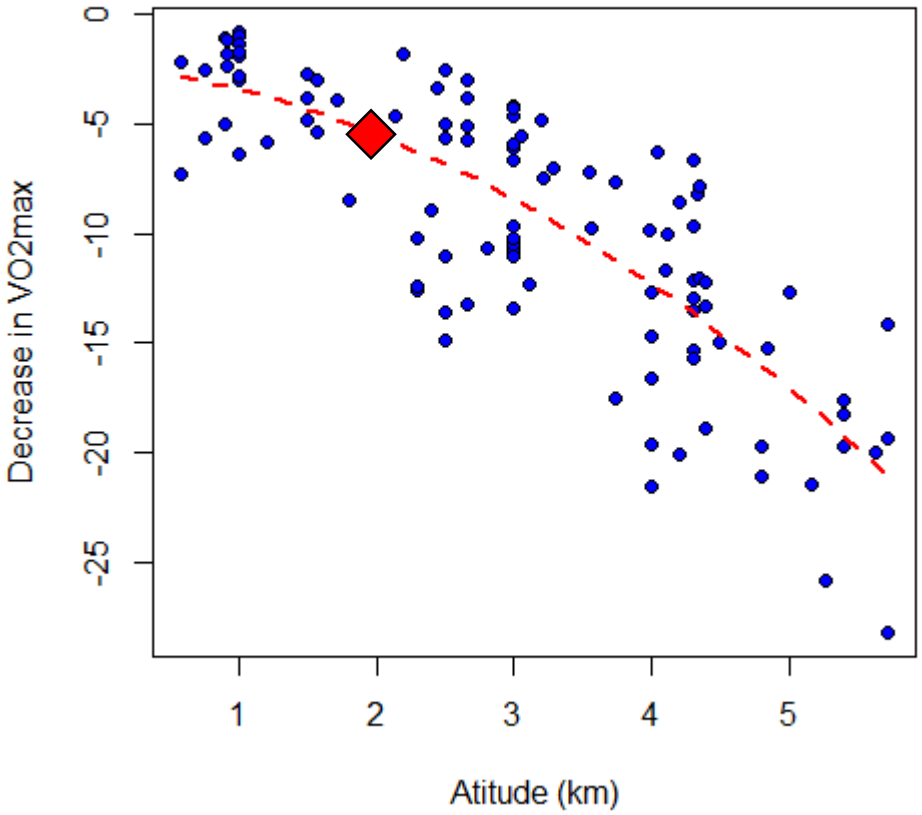
$$y_i = -2.2708 - 0.6921(ALT_i) - 0.4532\left(ALT_i^2\right) + e_i$$

$$y_i = -2.2708 - 0.6921(\mathbf{2}) - 0.4532(\mathbf{4}) + e_i$$

$$y_i = -5.4678 + e_i$$



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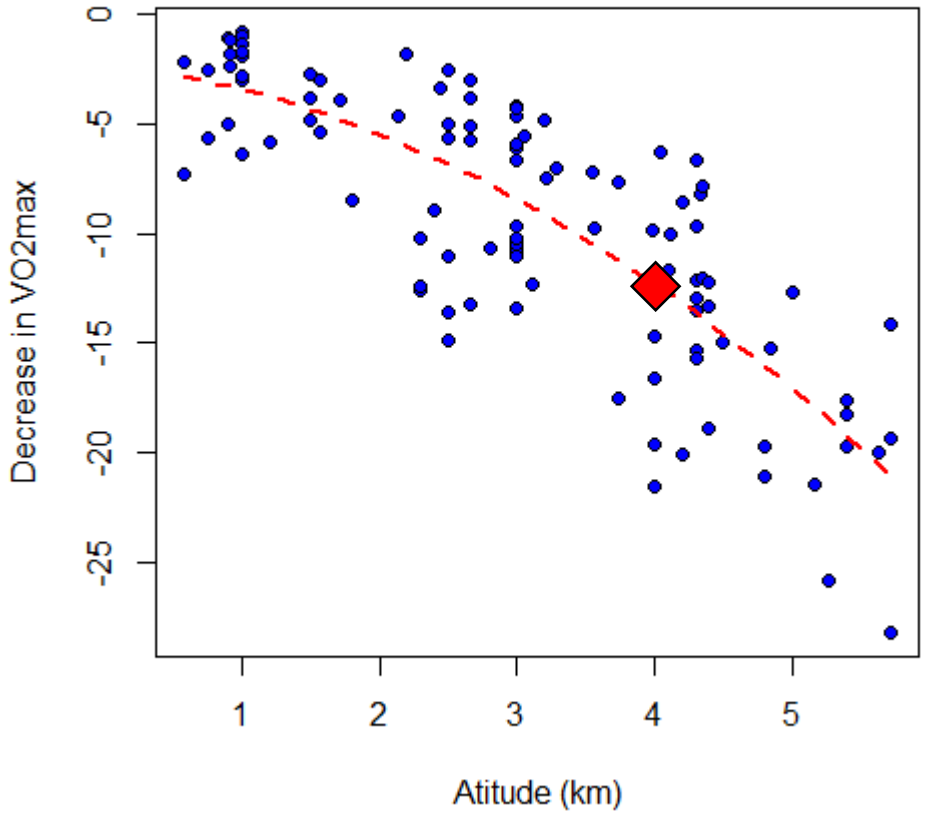
$$y_i = -2.2708 - 0.6921(\mathbf{4}) - 0.4532(\mathbf{16}) + e_i$$

$$y_i = -12.2904 + e_i$$



ANOVA table for the full model

Source	SSE	Df	MSE	F_obs	Pr(>F_obs)
Intercept	29.84	1	29.84	2.195	0.141
Alt (km)	5.39	1	5.39	0.396	0.530
Alt^2 (km^2)	89.59	1	89.59	6.589	0.012
Residuals	1386.74	102	13.59		
Total	1511.56	104			